

## Compare and contrast Mitosis and Meiosis Worksheet

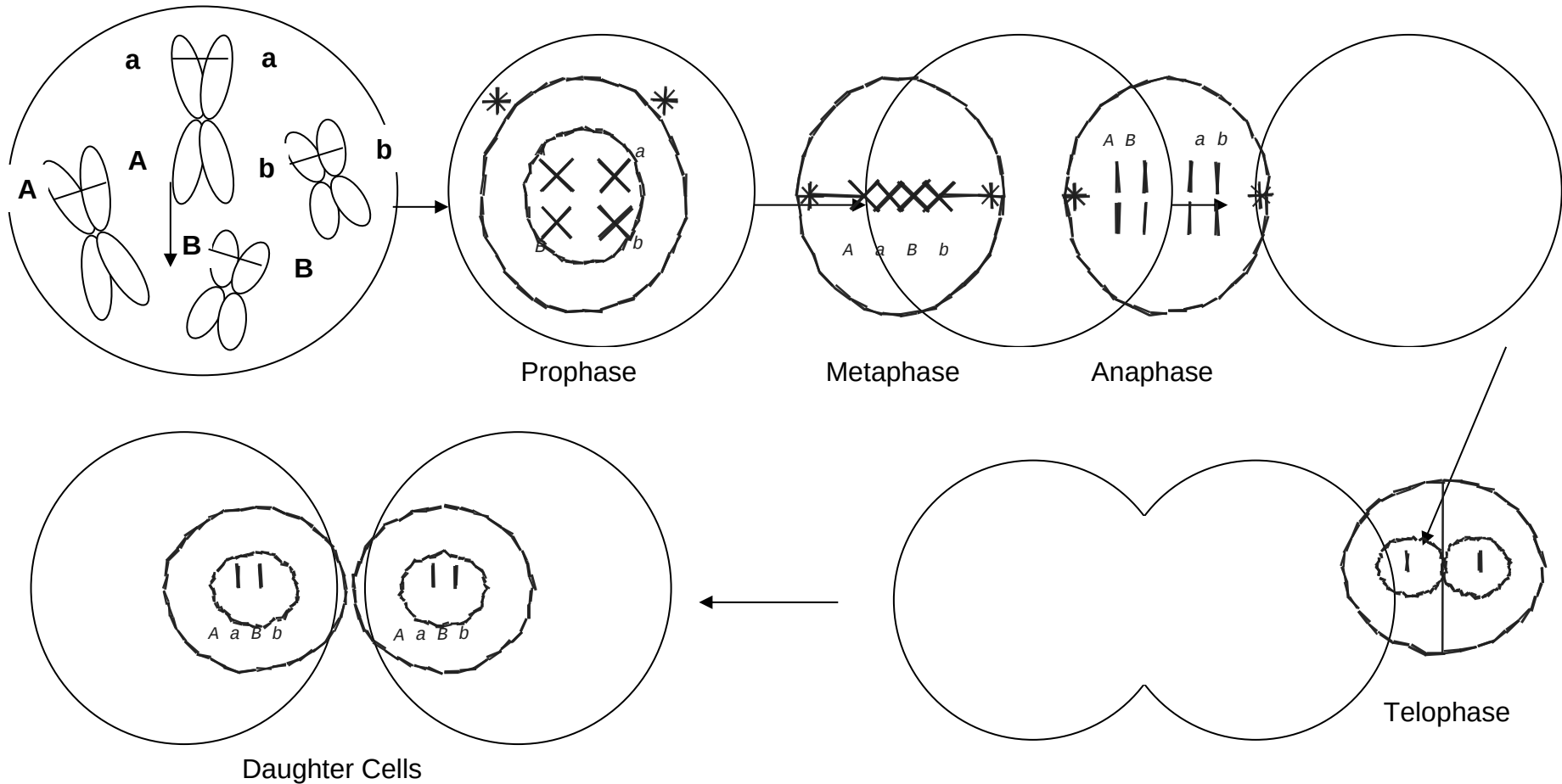
Part I Mitosis→ Use Diploid skin cells for Mitosis with  $2n = 4$  (as shown in picture below)

Material:

colored pencils

- blue and purple for dad's chromosomes (genes A, B)
- red and green from mom's chromosomes (genes a, b)

- I. Draw the chromosomes in the cell as it undergoes **Mitosis**:  
**Remember homologous chromosomes line up on separate spindle fiber**



## Part II Meiosis

II. Draw the chromosomes in the cell as it undergoes Meiosis (same parent cell):

**Use Use Diploid testical cells for Mitosis with  $2n = 4$  (as shown in the picture below)**

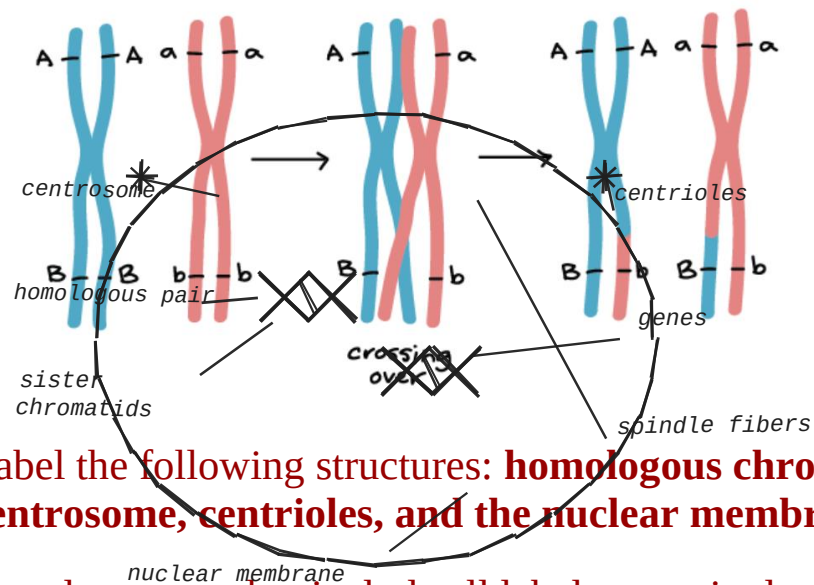
**Remember Homologous chromosomes line up on separate spindle fibers**

Material:

colored pencils

- blue and purple for dad's chromosomes (genes A, B)
- red and green from mom's chromosomes (genes a, b)

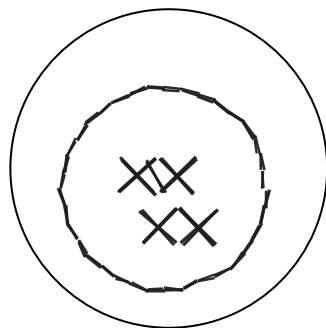
Show crossing over by switching the genes during Prophase I of meiosis



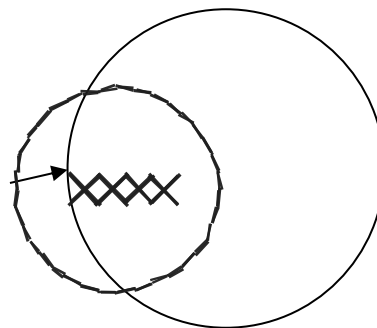
Note that chromosomes A, a are on one homologous pair of chromosomes in our exercise, and B, b are on a separate second homologous pair of chromosomes

Label the following structures: **homologous chromosomes, sister chromatids, genes, spindle fibers, centrosome, centrioles, and the nuclear membrane.**

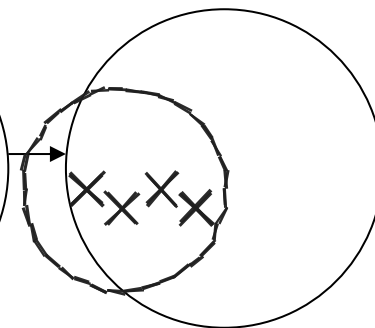
You do **not** need to include all labels on a single cell—spread them across the diagram so each structure is clearly visible and easy to identify.



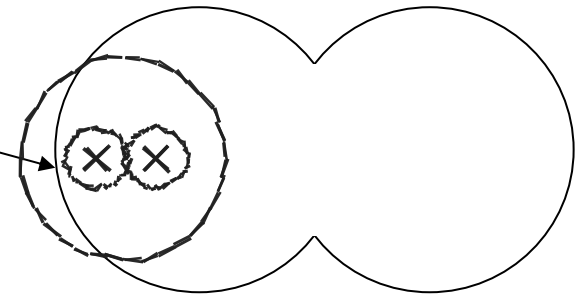
Prophase I



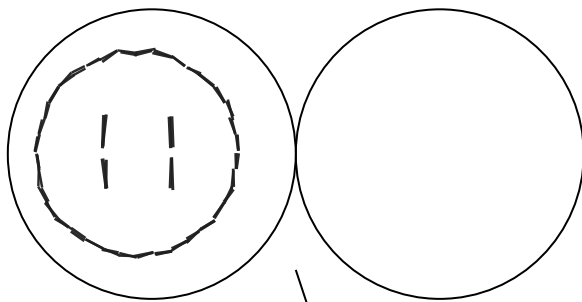
Metaphase I



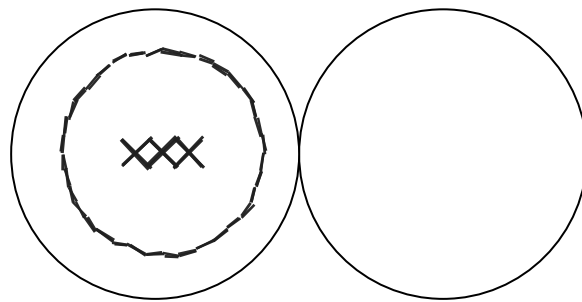
Anaphase I



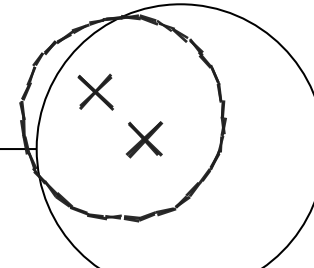
Telophase I



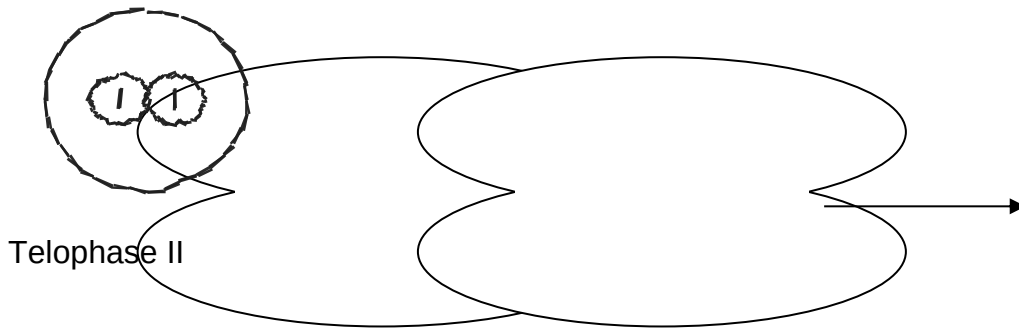
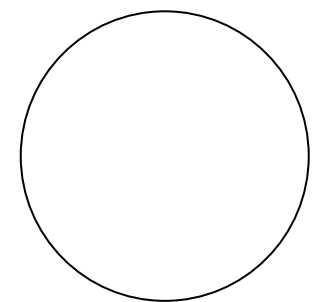
Anaphase II



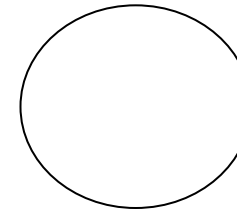
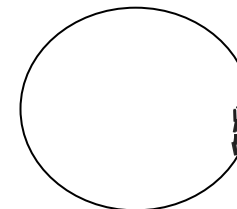
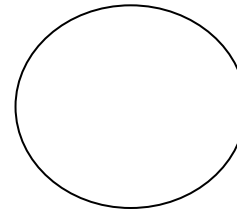
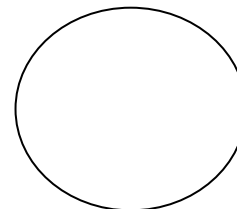
Metaphase II



Prophase II



Telophase II



4 haploid

Daughter Cells

### Part III: Complete the summary chart Comparing and Contrasting Mitosis and Meiosis

| Cell Processes       | Mitosis                                              | Meiosis                                              |
|----------------------|------------------------------------------------------|------------------------------------------------------|
| Creates              | <i>body cells</i>                                    | <i>sex cells<br/>(gametes)</i>                       |
| Definition           | <i>1 division -&gt;<br/>2 identical<br/>2n cells</i> | <i>2 divisions -&gt;<br/>4 different<br/>n cells</i> |
| End Products         | <i>2 diploid<br/>cells</i>                           | <i>4 haploid<br/>cells</i>                           |
| Steps                | <i>PMAT once</i>                                     | <i>PMAT I +<br/>PMAT II</i>                          |
| Type of Reproduction | <i>asexual</i>                                       | <i>sexual</i>                                        |

|                                                                                                                    |                                                              |                                                                   |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------|
| Are they identical to the parent cell?                                                                             | <i>yes</i>                                                   | <i>no</i>                                                         |
| When does cytokinesis occur?                                                                                       | <i>once<br/>(after telophase)</i>                            | <i>twice<br/>(after I and II)</i>                                 |
| How many times does the parent cell divide?                                                                        | <i>1 time</i>                                                | <i>2 times</i>                                                    |
| What happens to the number of chromosomes at the end of each process? Are they in pairs or individual chromosomes? | <i>stays <math>2n</math><br/>(46 in humans)<br/>in pairs</i> | <i>halves to <math>n</math><br/>(23 in humans)<br/>individual</i> |
| Why is each important?                                                                                             | <i>growth +<br/>repair</i>                                   | <i>variation +<br/>proper # for<br/>fertilization</i>             |

|                                                                                                       |                                                                   |  |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|--|
| How many chromosomes do human body cells and human sex cells have after they go through each process? | <i>Mitosis: body cells = 46</i><br><i>Meiosis: sex cells = 23</i> |  |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|--|

### Part III

Explain **nondisjunction**—when chromosomes or chromatids fail to separate properly—and how it can lead to conditions such as **Down syndrome** and **Turner syndrome**.

Then explain how these chromosomal abnormalities result in different physical and developmental effects.

*Nondisjunction = chromosomes do not separate right.*

*In meiosis I homologous chromosomes can fail to split;  
meiosis II sister chromatids can fail to split.*

*This makes gametes with extra or missing chromosomes.*

*Down syndrome: trisomy 21 (47 total).*

*Turner syndrome: monosomy X (45, X).*

*These change gene dosage, so development can differ:*

*Down can include learning delays and heart risks,*

*Turner can include short stature and ovarian issues.*